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Statistical Inference

Short notes

Markov's Inequality

$X \sim$ Random #; then for any $a > 0$

$$P\{X > a\} \leq \frac{E[X]}{a}$$

Chebyshev's Inequality

$X \sim N(\mu, \sigma^2)$ then $P\{|X-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}$

If $\text{var}(X) = 0$, then $P\{X = E[X]\} = 1$

Weak law of large #s:

$X_1, \ldots, X_n \sim$ Random sample $E[X_i] = \mu$,
then for any $\varepsilon > 0$,

$$P\left\{\left|\frac{X_1 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0$$

as $n \to \infty$

Sample mean

$$\bar{X} = \frac{X_1 + \cdots + X_n}{n} = \frac{\sum X_i}{n}$$

$$E[\bar{X}] = \mu, \quad \text{var}(\bar{X}) = \frac{\sigma^2}{n}$$

CLT

Sum of a large # of independent rvs has a distribⁿ that is approx. normal.

When we are approx. the sum of discrete rvs as random, we apply continuity corrections (0.5 extra on the either sides).

Approx distribution of sample mean

$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ if n is large using CLT

$$\Rightarrow \frac{\bar{X} - \mu}{\sigma^2/n} \sim Z \text{ or } N(0,1)$$

Strong law of large #s

If $X_1, \ldots, X_n$ is a sequence of independent trails, $E[X_i] = \mu$,
(or a random sample)
then with probability 1,

$$\frac{X_1 + \cdots + X_n}{n} \to \mu \quad \text{as} \quad n \to \infty$$

One-sided Chebyshev inequality

$$P\{X - \mu \geq a\} \leq \frac{\sigma^2}{\sigma^2 + a^2} \qquad a \geq 0$$

$$P\{X - \mu \leq -a\} \leq \frac{\sigma^2}{\sigma^2 + a^2}$$

$X : E[X] = \mu$
$\text{var}(X) = \sigma^2$

Sample variance

$$S^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{(n-1)}$$

$$E[S^2] = \sigma^2$$

If we sample from a normal population,

$$x_1, \ldots, x_n \sim N(\mu, \sigma^2)$$

$$\bar{x} \sim N(\mu, \sigma^2);$$

$$\frac{\bar{x}-\mu}{\sigma^2/\sqrt{n}} \sim Z \; ; \quad (n-1)\frac{s^2}{\sigma^2} \sim \chi^2_{n-1} \; ; \quad \frac{(\bar{x}-\mu)}{s/\sqrt{n}} \sim t_{n-1} \; ;$$

Point estimation

We want to estimate a parameter θ of a distribution.

$$\hat{\theta} = h(x_1, \ldots, x_n)$$

Evaluating estimators

Bias: $B(\hat{\theta}) = E[\hat{\theta}] - \theta$ If $B(\hat{\theta}) = 0$, then $\hat{\theta}$ is an unbiased estimator.

Mean squared error: $MSE(\hat{\theta}) = E[(\hat{\theta}-\theta)^2]$

$$= Var(\hat{\theta}) + Bias(\hat{\theta})^2$$

Consistency:

$\hat{\theta}$ is a consistent estimator of θ if $\lim_{n\to\infty} P[|\hat{\theta}-\theta| \geq \varepsilon] \to 0 \quad \forall \varepsilon > 0$

Maximum likelihood estimators

MLE for Bernoulli(p) parameter: $\hat{p} = \frac{\sum_{i=1}^{n} x_i}{n}$

for Binomial(m, θ) : $\hat{\theta} = \frac{\sum_{i=1}^{n} x_i}{m \cdot n}$

for Exp(θ) : $\hat{\theta} = \frac{1}{\frac{\sum_{i=1}^{n} x_i}{n}} = \frac{n}{\sum_{i=1}^{n} x_i}$

for $U(0, \theta)$: $\hat{\theta} = \max(x_1, x_2, \ldots, x_n)$

for poisson$(\lambda = \theta)$: $\hat{\theta} = \frac{\sum_{i=1}^{n} x_i}{n}$

for $\mu(\mu, \sigma^2)$: $\hat{\mu} = \frac{\sum_{i=1}^{n} x_i}{n}$ $\quad \hat{\sigma}^2 = \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{n}$

Confidence intervals

$P\{\hat{\theta}_1 < \theta < \hat{\theta}_n\} \geq (1-\alpha)$ with $(1-\alpha)\cdot 100\%$ confidence.

> $z_\alpha = \varphi^{-1}(1-\alpha)$ as $P\{Z > z_\alpha\} = \alpha$ and $z_{1-\alpha} = -z_\alpha$

> $t_{\alpha,n}$: $P\{T_n > t_{\alpha,n}\} = \alpha$ $\quad t_{1-\alpha} = -t_\alpha$

> $\chi^2_{\alpha,n}$: $P\{\chi^2_n > \chi^2_{\alpha,n}\} = \alpha$

(1-α) confidence interval for μ

$$\mu \in \left[\bar{x} - z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right] \quad \text{with } (1-\alpha)\cdot 100\% \text{ conf.}$$

One-sided:

$$\mu \in \left[\bar{x} - z_\alpha\frac{\sigma}{\sqrt{n}},\ \infty\right) \quad \text{with } (1-\alpha)\cdot 100\% \text{ conf.}$$

$$\mu \in \left(-\infty,\ \bar{x} + z_\alpha\frac{\sigma}{\sqrt{n}}\right]$$

(1-α) confidence interval for μ if σ is unknown

> find an upper limit for σ, $\sigma \leq \sigma_{max}$

> use S^2 as an estimator for σ^2

Confidence intervals for normal samples

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim Z, \qquad (n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$$

If $X_1, X_2, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$, $\bar{x} = \frac{\sum_{i=1}^n}{n}$, $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$, then

Parameter	Assumption	$(1-\alpha)\times 100\%$ C.I.	Lower Interval	Upper Interval
μ	σ^2 is known	$[\bar{x} - z_{\alpha/2}(\frac{\sigma}{\sqrt{n}}), \bar{x} + z_{\alpha/2}(\frac{\sigma}{\sqrt{n}})]$	$(-\infty, \bar{x} + z_\alpha(\frac{\sigma}{\sqrt{n}})]$	$[\bar{x} - z_\alpha(\frac{\sigma}{\sqrt{n}}), \infty)$
μ	σ^2 is unknown	$[\bar{x} - t_{\alpha/2,n-1}(\frac{s}{\sqrt{n}}), \bar{x} + t_{\alpha/2,n-1}(\frac{s}{\sqrt{n}})]$	$(-\infty, \bar{x} + t_{\alpha,n-1}(\frac{s}{\sqrt{n}})]$	$[\bar{x} - t_{\alpha,n-1}(\frac{s}{\sqrt{n}}), \infty)$
σ^2	μ is unknown	$[\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}]$	$(-\infty, \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}}]$	$[\frac{(n-1)s^2}{\chi^2_{\alpha,n-1}}, \infty)$

Prediction intervals

$X_1, \cdots, X_n, X_{n+1}$

Given the sample from 1 to n, we want X_{n+1}.

$$X_{n+1} \in \left[\bar{x}_n - t_{\frac{\alpha}{2},n-1}\cdot S_n\sqrt{1+\frac{1}{n}},\ \bar{X}_n - t_{\frac{\alpha}{2},n-1}\ S_n\sqrt{1+\frac{1}{n}}\right] \quad \text{with } (1-\alpha)\cdot 100\% \text{ conf.}$$

Estimating the difference in means of 2 different normal populations

$X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$

> If σ_1^2 and σ_2^2 are known,

$$\mu_1 - \mu_2 \in \left(\bar{x} - \bar{y} - z_{\frac{\alpha}{2}}\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}},\ \bar{x} - \bar{y} + z_{\frac{\alpha}{2}}\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}\right)$$

> If σ_1^2, σ_2^2 are unknown but equal,

$$S_p^2 = \frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2} \quad ; \quad \frac{\bar{x}-\bar{y}-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}} \sim t_{n+m-2}$$

$$\mu_1-\mu_2 \in \bar{x}-\bar{y} \pm t_{\frac{\alpha}{2},n+m-2} \cdot S_p\sqrt{\frac{1}{n}+\frac{1}{m}}$$

> If σ_1^2 and σ_2^2 are unknown and unequal, we can get an approx. C.I.

$$\mu_1-\mu_2 \in \left[\bar{x}-\bar{y} - z_{\frac{\alpha}{2}}\sqrt{\frac{S_1^2}{n}+\frac{S_2^2}{m}} \;,\; \bar{x}-\bar{y} + z_{\frac{\alpha}{2}}\sqrt{\frac{S_1^2}{n}+\frac{S_2^2}{m}}\right]$$

Approx C.I. for mean of Bernoulli r.v.

$$p \in \left[\hat{p} - z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \;,\; \hat{p} + z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right] \qquad \hat{p} = \text{sample mean}$$

Approx C.I. for mean of Exp r.v.*

$$x_1, \ldots, x_n \sim Exp\left(\frac{1}{\theta}\right)$$

$$\theta \in \left[\frac{2\bar{x}n}{\chi^2_{\frac{\alpha}{2},2n}} \;,\; \frac{2\bar{x}n}{\chi^2_{1-\frac{\alpha}{2},2n}}\right]$$

Hypothesis testing